Hexadecimal -- Guided Notes ANSWER KEY

Unit tp-1.3 | CompTIA Network+ N10-009 Obj. FC0-U71 1.2 | N10-008 FC0-U71 1.2

For instructor use / distribute after students complete the notes.

Question 1

Hexadecimal is a base-_______ number system. It uses the digits 0 through 9 and the letters _______ through _______. The letter A represents the decimal value _______, and the letter F represents _______.

Answer: 16; A through F; A=10, F=15.

Question 2

Convert the hexadecimal value 2F to decimal. Show your work:

2 is in the 16^1 position = 2 x _______ = _______

F (15) is in the 16^0 position = 15 x _______ = _______

Total = _______

Answer: 2 x 16 = 32; F(15) x 1 = 15; Total = 47.

Question 3

To convert binary to hexadecimal, group binary digits into sets of _______ starting from the right.

Convert the binary number 11010110:

Left group: _______ in binary = _______ in hex

Right group: _______ in binary = _______ in hex

Result: _______

Answer: 4 bits. Left group: 1101 = 13 = D. Right group: 0110 = 6. Result: D6 (or 0xD6).

Question 4

The prefix _______ is used in programming to indicate a hexadecimal value (e.g., 0xFF). The hex value 0xFF equals _______ in decimal, which in binary is _______ -- the maximum value of one _______.

Answer: 0x; 255; 11111111; byte.

Question 5

Hex is used for _______ addresses, which are 48-bit hardware identifiers written as six pairs of hex digits (e.g., AA:BB:CC:DD:EE:FF). Each pair represents _______ bits. Hex is also used for _______ codes in design (e.g., #FF5733).

Answer: MAC; 8 bits (one byte); color (HTML/CSS).

Question 6

Hex is also used for _______ addresses, where the location of data in RAM is expressed as a hex value. Programmers prefer hex over binary because it is more _______ while still mapping cleanly to binary -- every hex digit represents exactly _______ binary bits.

Answer: Memory; compact (readable); 4.

Question 7

Convert the decimal number 255 to hexadecimal. Show your work:

255 divided by 16 = ________ remainder ________

The remainder ________ is the rightmost hex digit = ________

The quotient ________ is the next digit

Result: ________

Answer: 255 / 16 = 15 remainder 15. Remainder 15 = F. Quotient 15 = F. Result: FF (or 0xFF).

Question 8 -- Real World Application

REAL WORLD: A network technician reads a system log and sees a memory address listed as 0x1A4F. A new employee asks why the address is not just written in decimal. Explain why engineers use hexadecimal instead of decimal or binary for memory addresses, using at least two reasons.

Answer: Reason 1 -- Compact representation of binary: each hex digit maps exactly to 4 binary bits, so 0x1A4F represents 0001 1010 0100 1111 in binary. Writing that as binary would require 16 digits; hex condenses it to 4 characters. Reason 2 -- Easier to read and transcribe than binary: humans make more errors copying long binary strings. Hex reduces cognitive load. Reason 3 -- Clean byte boundaries: since 1 byte = 8 bits = 2 hex digits, hex makes it easy to identify individual bytes in an address, which is useful when diagnosing memory and data alignment issues.